

Watching Two Boxes Talk

A step-by-step guide to recording the private conversation between a mixing desk and its stagebox — using a laptop, two network cables, and one free piece of software. No specialist network gear required. This edition covers bridging on macOS.

NO TAP NEEDED

NO MIRROR SWITCH NEEDED

~20 MINUTES

BENCH SETUP ONLY

THIS EDITION:

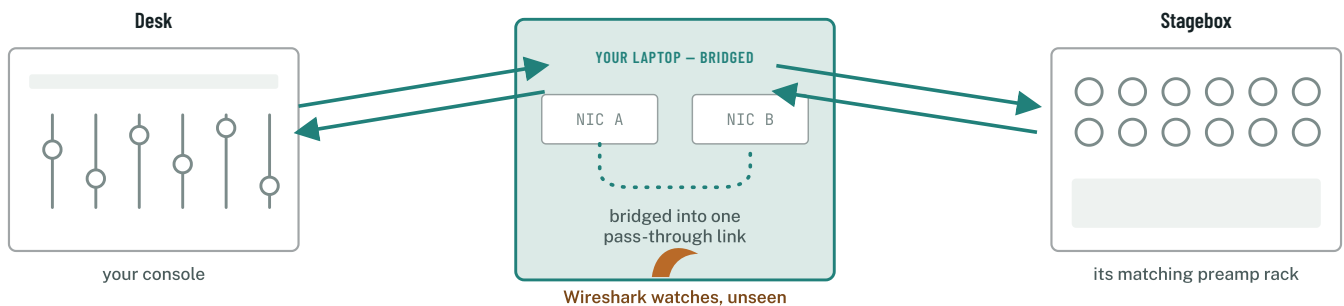
Windows

macOS

Linux

DIAGRAM — THE IDEA IN ONE PICTURE

01



Every packet that would normally jump straight from desk to stagebox now physically passes through your laptop first — like fitting a clear length of pipe into a hose.

The whole trick: sit your laptop *in* the cable run, tell its two network ports to quietly pass everything through, and let Wireshark watch what goes by.

Every desk-and-stagebox pair from the same manufacturer has a private handshake — how they find each other, pair up, and agree on gain and phantom power — that's never

been written down publicly. To build tools that can join that conversation from outside, we first have to record it happening between two devices that already trust each other.



Bench test, not showtime

Do this on a spare desk-and-stagebox pair, before or after a show — never on a live network mid-event. Inserting anything into a production Dante network, even briefly, isn't worth the risk.

§1 **What you'll need**

Nothing here is specialist. If you already run Dante gear, you likely own everything but the adapter cable — and that's a few pounds at any computer shop.

 The desk Your console. REQUIRED	 Its matching stagebox Same brand and family as the desk — you want the pairing they already trust. REQUIRED	 Two Ethernet cables The ordinary ones you'd use anyway. REQUIRED
 A laptop with two network ports A built-in Ethernet port (or Thunderbolt adapter) plus a cheap USB-to-Ethernet adapter covers almost any Mac. REQUIRED	 Wireshark, installed Free at wireshark.org. The installer also installs ChmodBPF — say yes when it asks, or capture will need sudo every time. REQUIRED	 A mirroring switch or tap Only if you already own one — see the alternatives near the end. Not required. OPTIONAL

§2 The setup, step by step

Nine steps, done in order, from cabling to a saved file ready to hand over.

1

Wire it up

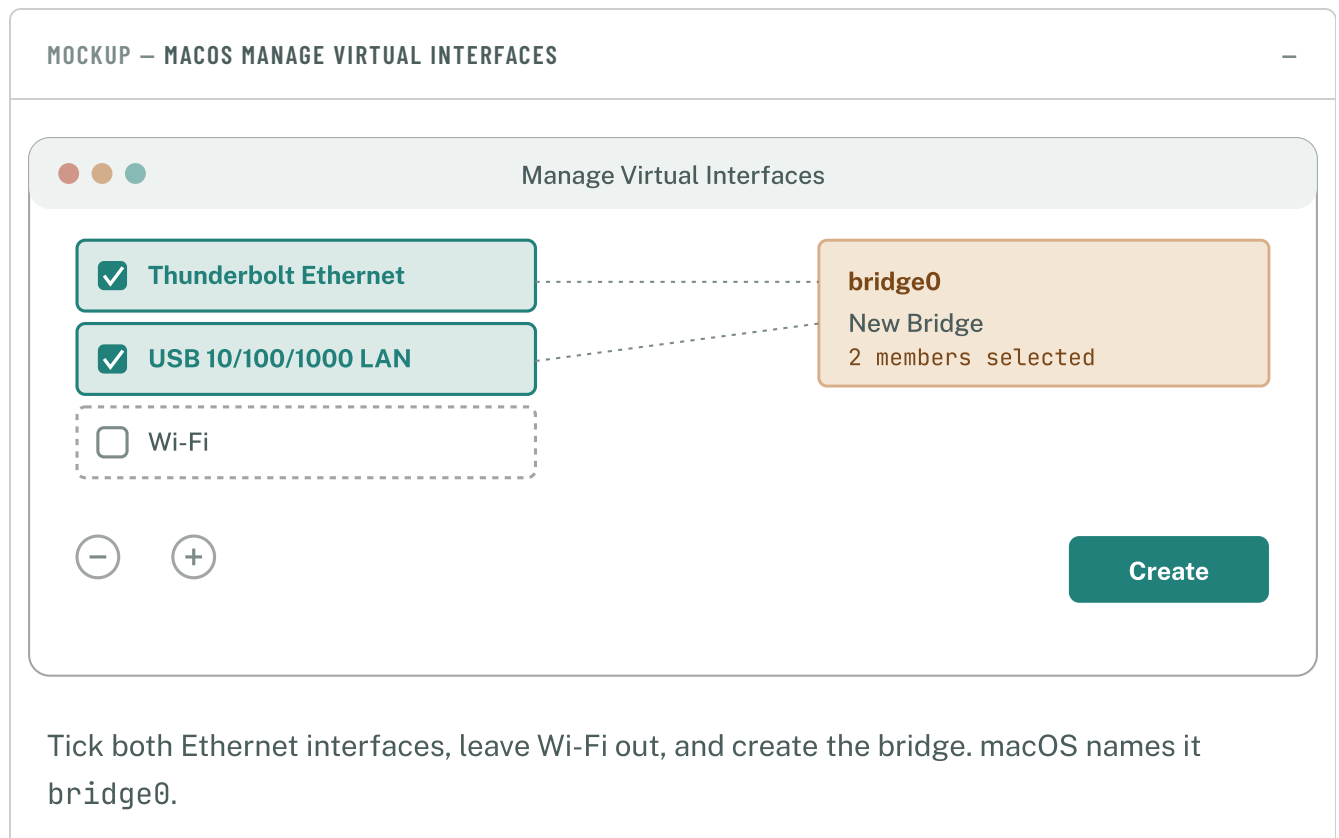
Run a cable from the desk's Ethernet port to **NIC A** on your laptop. Run a second cable from **NIC B** on your laptop to the stagebox's Ethernet port. That's it — no switch in between, your laptop *is* the connection now.

2

Bridge your two network ports

macOS can do this too — it's just tucked a little deeper than Windows, in a menu most people never open. Open `System Settings → Network`, click the `...` button at the bottom of the interface list (a gear icon on older macOS) and choose `Manage Virtual Interfaces...`.

Click `+ → New Bridge`, tick both of your Ethernet interfaces — the built-in port and your USB-to-Ethernet adapter — and click `Create`.



Using Windows or Linux instead? There's a matching guide for each — the bridging step works differently, but everything else here is the same.

3

Open Wireshark and pick your interface

Look for bridge0 in Wireshark's interface list — that's the one to capture on.

MOCKUP — WIRESHARK INTERFACE LIST

The Wireshark Network Analyzer

bridge0



← capture here

en0 — Thunderbolt Ethernet

idle

en5 — USB 10/100/1000 LAN

idle

en1 — Wi-Fi

idle

Nothing on it yet — that's expected before anything's powered on.

If nothing shows up on bridge0 once traffic is flowing, select both physical interfaces together instead and start both — Wireshark merges them into one capture automatically.

4

Start the capture first

Click the blue shark-fin Start button *before* powering anything on. This is what catches the very first discovery packets — the part we're actually missing today.

5

Power everything on

Turn the stagebox and desk on in whatever order you'd normally use, then wait about 30 seconds for them to find each other over the network.

6

Pair them, like normal

From the desk's own screen, connect to the stagebox exactly as you always would. No special settings — routine operation is exactly what we want recorded.

7

Wiggle a few things

Once, each: nudge a gain knob physically on the stagebox; nudge a different one from the desk's on-screen control; toggle phantom power on a channel; then disconnect the desk from the stagebox in software and reconnect it. That re-pairing moment is one of the most useful bits in the whole capture.

8

Stop and save

Hit the red square, then . Give it a name that says what it is — something like `q11_tio1608_2026-07-14.pcapng` — and save.

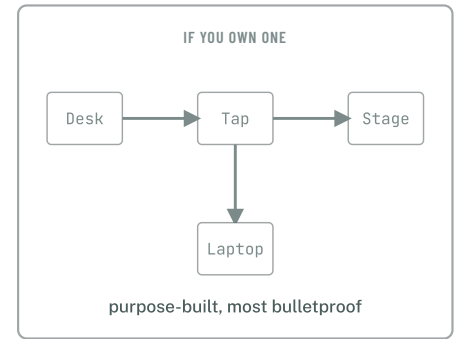
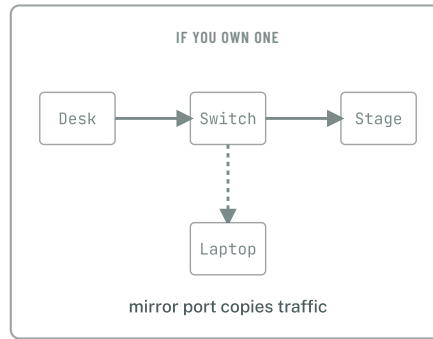
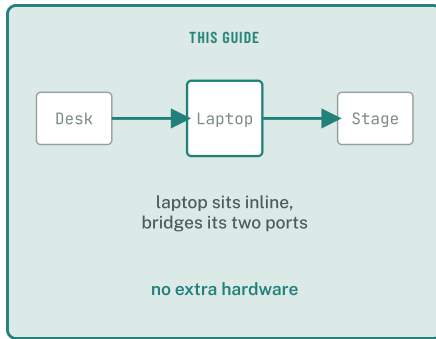
9

Send it over, then tidy up

The `.pcapng` file on its own is enough — no need to trim or export anything yourself. Afterward, undo the bridge: , select `bridge0`, and click the button to delete it.

§3 Already own a mirroring switch or a tap?

Neither is necessary — the laptop-bridge method above works fine on its own — but if you already have one of these, it's a perfectly good alternative to bridging.



Managed switch with a mirror / SPAN port

Plug the desk, stagebox, and your laptop into the switch as normal, then configure one port to mirror the desk and stagebox's traffic onto your laptop's port. Capture directly on that single NIC — no bridging needed.

NEEDS ADMIN ACCESS TO THE SWITCH

Hardware network tap

A small dedicated box that sits inline between desk and stagebox, just like the laptop-bridge method, and quietly copies every packet out a third port to your laptop. The most bulletproof option, since it never routes the devices' whole conversation through a general-purpose computer.

EXTRA KIT TO BUY — NOT REQUIRED

§4 If something's not working

? I can't find "Manage Virtual Interfaces..."

It's under the `...` (or gear) button at the very bottom of the Network settings interface list — on older macOS versions it's a dropdown directly under the gear icon rather than a separate button.

? Wireshark asks for my password, or the interface list is empty

The ChmodBPF helper wasn't installed. Re-run the Wireshark .pkg installer and make sure the ChmodBPF step is ticked, then log out and back in.

? bridge0 shows no traffic

Capture on both individual physical interfaces at the same time instead — Wireshark merges them into a single, correctly-ordered capture automatically.

? The desk and stagebox won't find each other through the laptop

Check `System Settings → Network → Firewall` isn't blocking the multicast traffic discovery relies on, and try turning Wi-Fi off entirely so it can't interfere.

Part of the Dante-BabelBox emulation research.

A finished .pcapng file is all we need — send it over as-is.